

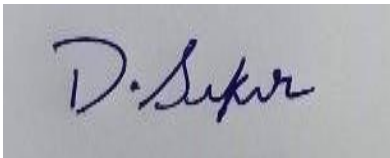
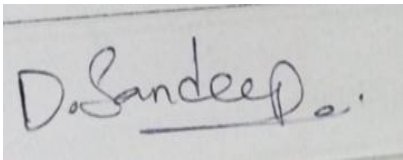
Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. TITLE: Smart Helmet with Health Monitoring

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

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3. DESCRIPTION OF THE INVENTION: The Smart Helmet with Health Monitoring is an advanced protective device designed to continuously observe the wearer's health parameters and detect potential medical emergencies. The helmet integrates multiple sensors capable of measuring vital indicators such as heart rate, blood pressure, body temperature, and brain activity, ensuring uninterrupted health surveillance during high-risk activities including motorcycle riding, cycling, and working in hazardous environments.

In addition to health tracking, the helmet is equipped with accelerometers and gyroscopes that identify sudden impacts, falls, or collisions. Upon detecting abnormal events, the system immediately alerts the wearer or designated emergency responders. The helmet also incorporates GPS technology that automatically transmits the user's real-time location to emergency services or pre-registered contacts when an accident or irregular health reading is detected. Machine learning techniques are applied to analyze the collected data, allowing the system to recognize potential health risks and improve predictive accuracy over time.

A. PROBLEM ADDRESSED BY THE INVENTION:

Road safety concerns have significantly increased, particularly for motorcyclists, alongside a growing number of health-related emergencies occurring during physically demanding activities. Conventional helmets provide physical head protection but lack integrated health monitoring capabilities, which are essential for early detection of accidents or medical issues.

Motorcyclists, cyclists, and individuals working in hazardous environments face a high risk of traumatic brain injuries, heart attacks, strokes, and other sudden health emergencies. Furthermore, there is an increasing demand for proactive health monitoring during activities involving intense physical exertion. Existing helmet solutions fail to deliver real-time health information to users or emergency responders, thereby increasing the likelihood of delayed medical intervention. A helmet integrated with real-time health monitoring can identify and communicate potential health threats to both the wearer and emergency services, significantly enhancing safety and increasing survival chances in critical situations.

B. OBJECTIVE OF THE INVENTION

The main objective of the Smart Helmet with Health Monitoring System is to improve user safety by integrating advanced health tracking and emergency response functionalities into a protective helmet. The specific objectives are outlined below:

1) Real-time Health Monitoring:

Continuous observation of body temperature, heart rate, and blood oxygen saturation levels to enable early identification of potential health abnormalities.

2) Enhanced Communication System:

Integration of Bluetooth and wireless connectivity to establish seamless interaction with a mobile application for real-time data access, alert notifications, and report generation.

3) Environmental Hazard Detection:

Incorporation of environmental sensors capable of detecting hazardous conditions such as extreme temperatures, toxic gases, and reduced oxygen levels, thereby preventing health risks.

4) Intelligent Warning System:

Provision of real-time audio alerts and visual notifications for abnormal health readings to immediately inform the user and prompt timely action.

C. STATE OF THE ART/ RESEARCH GAP/NOVELTY: Describe your invention fulfil the research gap?

Sr. No.	Patent I'd	Abstract	Research Gap	Novelty
1.	US20120172052A1	A headgear equipped with impact detection sensors to assess head-related injuries occurring during crashes serves as its main feature. This accessory incorporates an emergency alert function to notify medical responders after a car crash occurs.	The described patent includes impact detection however it does not contain the capability to monitor health indicators such as heart rate and temperature and oxygen levels. The patent design does not contain any components to prevent health risks or implement health protection protocols.	The proposed invention combines vital sign sensors in a comprehensive health monitoring system which tracks health continuously to provide preventive care and detect medical emergencies early. The device includes GPS monitoring capabilities to improve emergency response operations.
2.	US20190261543A1	The gadget gives motorcycle riders a smart helmet with GPS tracking capabilities along with accident detection technology. A smart helmet system offers GPS tracking together with accident detection features that will send emergency service locations to users following a crash event.	The patented automation system does not contain environmental hazard detection features which are essential for industrial and construction site operations.	The smart helmet employs environmental sensors to identify dangerous surroundings which improves workplace safety in construction sites and mining operations alongside other risky environments.

3.	US20170293145A1	The device provides health monitoring with heart rate and body temperature tracking through wireless transmission of data.	The separate design of this health monitoring device creates barriers for motorcyclists and industrial workers because they need free use of their hands while wearing protective helmets.	The invention implements health monitoring sensors into a single-dominant helmet structure offering safe convenience alongside real-time alert systems through a unified system without wearable accessories.
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D. DETAILED DESCRIPTION:

1. Health Monitoring and Safety Support

What It Does:

The Smart Helmet with Health Monitoring System is designed to provide continuous observation of essential health parameters, including heart rate, body temperature, and blood oxygen saturation levels. The system identifies irregular health readings and immediately alerts the wearer as well as predefined emergency contacts during critical situations.

In addition, impact detection sensors monitor accidents and instantly transmit the user's location to emergency services when falls, collisions, or severe impacts occur.

How It Works:

The helmet is equipped with embedded sensors that continuously collect real-time medical data. When sensor readings exceed safe threshold values, the system activates warning alerts and sends notifications through a connected mobile application. The integrated GPS module automatically shares the precise location of the user with emergency responders whenever an emergency event is detected.

2. Accident Detection and Emergency Response System

What It Does:

The helmet incorporates an accelerometer and a gyroscope to detect sudden impacts, collisions, or falls. When an accident is identified, the system automatically triggers emergency alerts and shares the user's exact geographic location with medical personnel or designated emergency contacts.

How It Works:

During an accident, the sensors detect abnormal motion patterns and initiate a countdown timer. If the user fails to respond within the defined time limit, the helmet automatically sends an SOS alert containing location details to emergency contacts, ensuring immediate assistance.

3. Environmental Hazard Detection System

What It Does:

The helmet includes environmental sensors that continuously monitor hazardous external conditions such as extreme temperatures, toxic gas exposure, and dangerously low oxygen levels. This system helps users avoid potentially life-threatening environments.

How It Works:

Sensors installed within the helmet track environmental parameters in real time. When unsafe conditions are detected, the system generates audible alerts and visual warnings, enabling users to take preventive action. This feature is especially beneficial for industrial workers, miners, and construction personnel operating in hazardous locations.

4. Smart Communication System

What It Does:

The helmet enables real-time data exchange between the device and a mobile application using Bluetooth and Wi-Fi connectivity. Users can monitor health metrics, configure emergency contacts, and receive instant alerts through the connected application.

How It Works:

Using wireless communication modules, the helmet connects to a dedicated smartphone application. The app interface allows users to view live health data, review historical incident records, and analyze environmental conditions collected during helmet usage.

5. Power Management System

What It Does:

The helmet is equipped with an efficient power management system designed to optimize energy consumption and extend operational time.

How It Works:

An automatic power-saving mode activates when no user interaction is detected. The rechargeable lithium-ion battery supports prolonged usage, making the helmet suitable for long rides, extended work shifts, and outdoor activities without frequent recharging.

6. User Data Logging and Analysis

What It Does:

The system continuously records health metrics, accident events, and environmental sensor data. This information provides users with meaningful insights into their overall health and safety conditions.

How It Works:

Collected data is securely stored within the helmet until it is synchronized with the mobile application. Through data analysis and visualization, users can observe health trends, identify patterns, and receive personalized safety recommendations.

7. Voice Alert and Notification System

What It Does:

The helmet integrates a built-in speaker system that delivers instant voice alerts regarding abnormal health readings, environmental dangers, and emergency situations.

How It Works:

When the system detects potential risks, it generates clear voice notifications to alert the user immediately. This feature ensures that warnings remain accessible even in situations where the user cannot access their mobile phone.

E. RESULTS AND ADVANTAGES:

The Smart Helmet with Health Monitoring System integrates modern safety technology with healthcare monitoring by combining multiple sensors, automated emergency assistance, and environmental tracking capabilities. By addressing critical limitations in conventional safety equipment, this system provides both preventive healthcare support and rapid emergency response mechanisms.

1. Enhanced Personal Safety and Emergency Response

The smart helmet delivers superior personal protection by incorporating impact detection sensors that automatically activate emergency alert mechanisms.

When the system detects an accident or a severe impact, it instantly notifies emergency contacts while transmitting the user's real-time GPS location.

This automated alert functionality reduces emergency response time, significantly increasing survival chances for motorcyclists, construction workers, and individuals operating in isolated or high-risk environments.

2. Continuous Health Monitoring for Early Intervention

The system provides users with real-time access to critical health indicators such as heart rate, body temperature, and blood oxygen levels.

Immediate alerts are generated when abnormal readings are detected, enabling users to take preventive action before conditions worsen.

This capability is particularly beneficial for individuals engaged in physically demanding occupations where health risks are higher.

3. Intelligent Environmental Hazard Detection

The helmet incorporates environmental sensors that continuously monitor exposure to hazardous gases, extreme temperatures, and reduced oxygen levels.

Both audio and visual alert systems notify users of unsafe environmental conditions, allowing timely preventive measures.

This feature makes the helmet especially suitable for use in high-risk locations such as construction sites, mining operations, and chemical processing facilities.

4. User-Centric Design for Maximum Comfort

Despite the integration of multiple sensors and electronic components, the helmet maintains a lightweight and ergonomically optimized structure.

The design ensures long-term comfort, allowing motorcyclists and industrial workers to wear the helmet for extended durations without discomfort.

Improved ventilation and airflow systems further enhance user comfort during prolonged usage.

5. Smart Data Logging and Analysis for Proactive Care

Health records, accident data, and environmental measurements are securely stored within the system.

The mobile application enables users to track long-term health trends, analyze patterns, and receive personalized insights.

Visual data representation tools help users better understand their health status, supporting informed decision-making.

6. Automated Emergency Assistance Without Manual Input

The helmet provides fully automated emergency assistance without requiring manual user intervention.

In critical situations, the system autonomously sends alerts to emergency contacts, ensuring rapid response even if the user is unconscious or unable to react.

This functionality is especially important for motorcyclists and workers operating in remote or isolated areas.

7. Multi-Device Integration for Seamless Connectivity

The system supports seamless connectivity with smartphones, smartwatches, and other IoT-enabled devices through Bluetooth and Wi-Fi technology.

Users can receive real-time alerts, monitor health data, and customize emergency contact settings across multiple connected devices.

Health data can also be transmitted wirelessly to family members, caregivers, or smart home systems for continuous monitoring.

8. Customizable Alert System for Enhanced User Control

Users are provided with the flexibility to configure personalized alert thresholds for heart rate, body temperature, and environmental hazards.

This customization enables tailored protection for individuals with pre-existing medical conditions or specific safety requirements.

9. Workplace Safety Enhancement for Industrial Use

The helmet is designed in compliance with industrial safety standards, making it suitable for hazardous occupational environments.

Continuous monitoring of worker health and environmental conditions helps reduce workplace accidents and improve employee safety.

Organizations benefit from improved safety compliance and enhanced workforce protection.

10. High Commercialization Potential

The Smart Helmet with Health Monitoring System demonstrates strong potential for commercialization across multiple industries.

Motorcycle safety gear manufacturers can integrate the system into premium helmet models to enhance product value.

Industrial safety solution providers can adopt the helmet to improve safety measures in hazardous work environments.

Healthcare providers can utilize the helmet to monitor high-risk patients, including elderly individuals and those with chronic health conditions.

11. Promotes Proactive Health Awareness

By continuously analyzing health data, the system encourages users to adopt preventive health practices.

The helmet provides proactive recommendations related to hydration, rest, and avoidance of unsafe conditions, supporting long-term well-being.

F. EXPANSION:

Expansion of Variables for Patent Coverage: Smart Helmet with Health Monitoring System

1. Sensor Data Input Variables

The selected variables define sensor placement, accuracy of data acquisition, and overall system performance.

Vital Sign Sensors:

The health monitoring system includes a heart rate sensor designed to detect variations in cardiovascular activity.

A body temperature sensor is used to measure thermal changes, allowing identification of overheating or hypothermic conditions.

A blood oxygen saturation (SpO₂) sensor monitors oxygen levels in the bloodstream, supporting respiratory health assessment.

Impact and Motion Sensors:

An accelerometer detects sudden shocks, rapid movements, falls, or abrupt head motion.

A gyroscope measures rotational movement, which is critical for assessing accident severity and angular impact dynamics.

Pressure sensors determine helmet fit and detect sudden pressure changes that may occur during collisions.

Environmental Sensors:

Gas sensors identify the presence of hazardous gases such as carbon monoxide, methane, or smoke.

Temperature sensors monitor heat exposure in industrial or outdoor environments.

Humidity sensors track moisture and condensation levels, helping to prevent discomfort and internal fog formation.

Sensor Placement Variables:

Sensors are embedded within the inner padding of the helmet to maintain proximity to vital arteries for accurate readings.

Impact sensors are strategically distributed across the outer shell to capture angle-specific collision data.

2. Machine Learning Algorithms

These variables incorporate artificial intelligence techniques that enhance system performance through adaptive learning and pattern recognition.

Supervised Learning Models:

Trained health datasets are used to identify indicators of heat exhaustion, dehydration, and abnormal heart conditions.

Unsupervised Learning Models:

The system analyzes data to detect unusual head movements, fatigue patterns, and irregular riding behavior without predefined labels.

Reinforcement Learning Models:

Alert sensitivity improves over time by learning from user feedback and environmental conditions, enabling adaptive response behavior.

Predictive Analytics Algorithms:

The system forecasts potential risks by recognizing correlations between prolonged fatigue, unsafe environments, and abnormal health parameters.

3. Helmet Design and Structural Parameters

Design variables focus on durability, comfort, and optimal performance.

Outer Shell Materials:

Materials such as polycarbonate, fiberglass, and carbon fiber are utilized for their high impact absorption capabilities.

Inner Lining Design:

Expanded polystyrene (EPS) foam is used for effective energy absorption during impacts.

Moisture-wicking fabrics enhance comfort by reducing sweat accumulation.

Airflow Management:

Integrated ventilation channels allow controlled airflow to regulate internal temperature and prevent heat buildup.

Ergonomic Design:

The helmet is engineered to evenly distribute weight, reducing neck strain and improving comfort during prolonged use.

4. Health Monitoring Algorithms and Thresholds

These variables define how the system evaluates health data and identifies critical medical conditions.

Threshold Variables for Vital Signs:

Customizable heart rate limits (for example, a normal range of 60–100 bpm).

Adjustable body temperature thresholds enable early detection of fever or heat-related stress.

Adaptive Alert Sensitivity:

Alert parameters automatically adjust based on user activity levels, ranging from resting states to intense physical activity.

Noise Filtering Algorithms:

Signal processing techniques differentiate between false readings caused by helmet movement and genuine health abnormalities.

5. Emergency Response and Notification System

Variables ensure effective emergency communication and response.

GPS Accuracy Variables:

Users can select location accuracy levels depending on whether they are in urban or rural environments.

Response Timing Variables:

Configurable delay timers allow brief response windows before automatic alerts are sent if the user remains unresponsive.

Communication Protocols:

Bluetooth, Wi-Fi, and GSM modules support multi-channel transmission of emergency alerts.

Multi-Level Emergency Contact System:

Alerts are issued sequentially, starting with close family members and escalating to medical services if required.

6. User Interface and Control Variables

These factors determine how users interact with the helmet system.

Voice Command Integration:

Hands-free voice commands allow users to perform health checks, trigger SOS alerts, and modify settings.

Mobile App Control:

The mobile application enables personalized dashboards for monitoring health data, managing contacts, and receiving alerts.

LED Notification System:

Integrated LED indicators provide visual warnings for environmental hazards and abnormal health readings.

7. Power Management and Energy Efficiency Variables

These variables manage power consumption and energy optimization.

Battery Type and Capacity:

The helmet uses a rechargeable lithium-ion battery capable of operating for approximately 12–15 hours.

Power-Saving Mode:

Automatic shutdown or reduced activity modes activate when the helmet is not in use to conserve energy.

Fast-Charging Support:

Fast-charging functionality minimizes downtime between usage cycles.

8. Data Privacy and Security Variables

These variables govern data protection and secure handling of user information.

Encryption Protocols:

All sensor data is encrypted during transmission to protect user privacy and prevent unauthorized access.

G. WORKING PROTOTYPE/ FORMULATION/DESIGN/COMPOSITION:The following is the Flowchart depicting the brief of the system

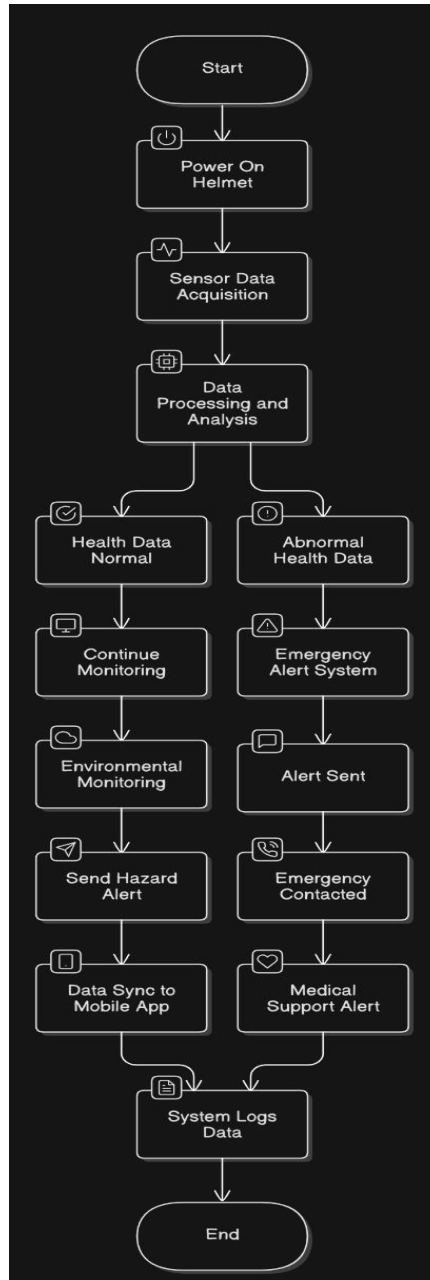


Figure 1 Flowchart for Smart Helmet With Health Monitoring System

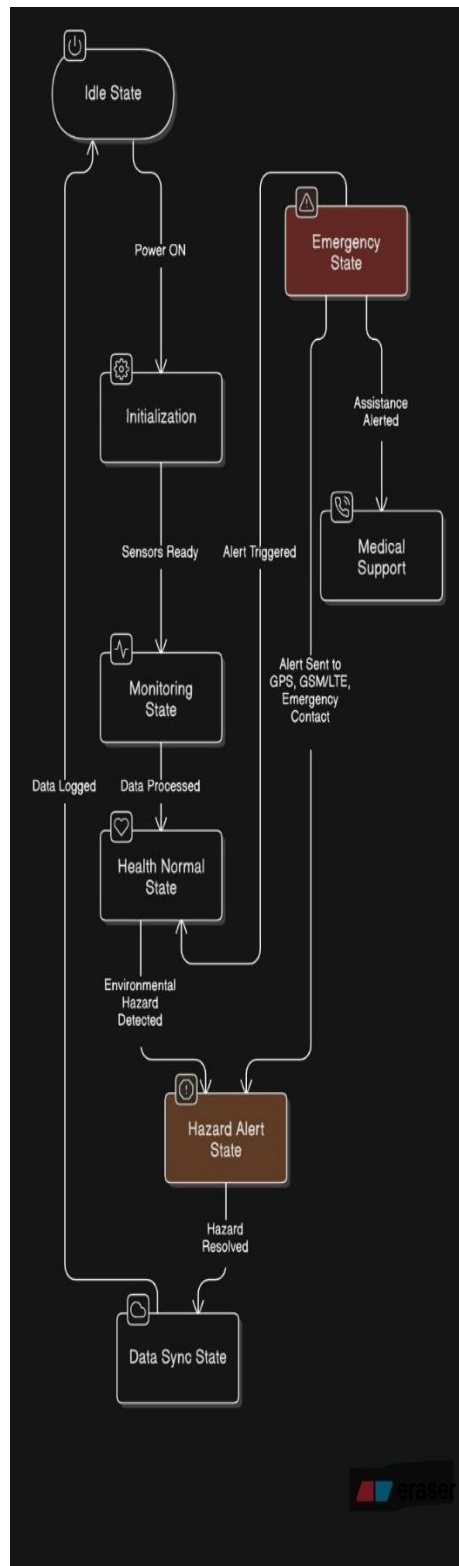


Figure 2 State chart for Smart Helmet With Health Monitoring System

The detailed description of the State Chart for Smart Helmet with Health Monitoring System is presented below.

1. Idle State (Helmet Powered OFF)

Description:

In this state, the helmet remains completely inactive when it is powered off.

All sensors, monitoring modules, and the Microcontroller Unit (MCU) are disabled, and no data collection or processing occurs during this phase.

Transition

Trigger:

The system transitions to the Initialization State when the user powers on the helmet.

2. Initialization State (Sensors, MCU Start)

Description:

Once powered on, the MCU initializes and activates all system components.

During this stage, sensors responsible for heart rate, body temperature, oxygen saturation, impact detection, and environmental monitoring are enabled and calibrated to ensure accurate data collection.

Transition

Trigger:

After successful calibration and readiness of all sensors, the system moves to the Monitoring State.

3. Monitoring State (Vital Signs, Impact, Environment Data)

Description:

In this operational state, the helmet continuously monitors the wearer's health parameters and surrounding environmental conditions in real time.

The system tracks:

- Heart rate
- Body temperature
- Oxygen saturation levels
- Impact or collision detection
- Environmental hazards such as carbon monoxide, toxic gases, and excessive heat

Transition

Trigger:

If all measured values remain within safe limits, the system transitions to the Health Normal State.

Detection of abnormal health data or environmental threats causes the system to move into either the Emergency State or the Hazard Alert State.

4. Health Normal State (Continue Monitoring)

Description:

This state confirms that all monitored health and environmental parameters are within predefined safety thresholds.

The helmet continues passive monitoring without triggering alerts.

Transition

Trigger:

Detection of hazardous environmental conditions initiates a transition to the Hazard Alert State.

Identification of abnormal health conditions causes the system to enter the Emergency State.

5. Emergency State (Alert Triggered)

Description:

The system enters the Emergency State when critical medical conditions or severe impacts are detected, including:

- Sudden reduction in oxygen levels
- Abnormal heart rate readings
- Strong head impact indicating a possible accident

During this state, alerts are transmitted using:

- GPS for real-time location tracking
- GSM/LTE networks for emergency messaging or calls

Emergency contacts are immediately notified about the user's condition and location.

Transition

Trigger:

After alerts are successfully transmitted, the system proceeds to the Medical Support Assistance State or returns to the Monitoring State if conditions stabilize.

6. Hazard Alert State (Toxic Gas / Impact Detected)

Description:

The helmet enters this state upon detecting dangerous environmental conditions in the surrounding area.

Alerts are issued for hazards such as:

- Toxic gas presence (e.g., carbon monoxide)
- Extreme temperature exposure
- Potential impacts or collision risks

The alert system prompts users to take immediate preventive actions.

Transition

Trigger:

Once hazardous conditions subside, the system transitions to the Data Sync State.

7. Medical Support Assistance State

Description:

In this state, the system directly communicates with medical support services when emergency intervention is required.

Notifications are shared with:

- Emergency responders, hospitals, and paramedics
- Family members or designated emergency contacts

Transition

Trigger:

After emergency services are notified, the system either resumes the Monitoring State or logs the event data for future analysis.

8. Data Sync State (Sync to Mobile App)

Description:

The helmet connects to the mobile application to synchronize collected data.

Emergency events, health records, and environmental data are stored securely, enabling performance evaluation and system improvement.

Transition

Trigger:

After successful data synchronization, the system transitions back to the Idle State.

9. Idle State (Helmet Powered OFF)

Description:

When the helmet is powered off, all system operations terminate safely.

The system shuts down active processes and remains idle until the next activation cycle.

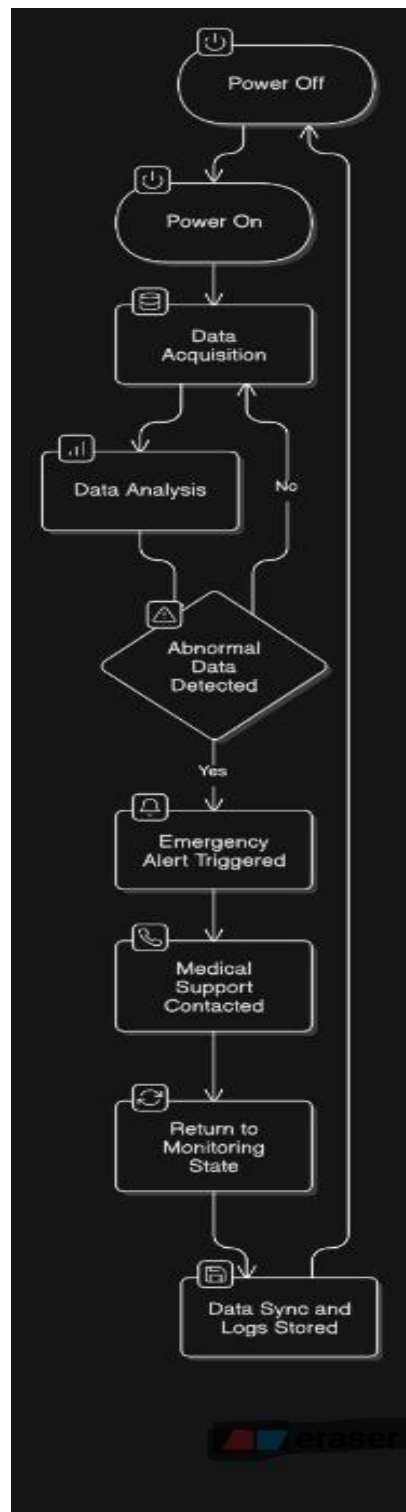


Figure 3 State Diagram for Smart Helmet With Health Monitoring System

Transition Flow:

The state diagram illustrates the sequential flow of operations between different system states, ensuring continuous monitoring, timely alert generation, and efficient data synchronization.

Transitions between states are initiated by real-time sensor inputs, anomaly detection mechanisms, and system feedback responses to enhance overall user safety.

G. EXISTING DATA: NA**4. USE AND DISCLOSURE (IMPORTANT):** Please answer the following questions:

A. Have you described or shown your invention/ design to anyone or in any conference?	YES ()	NO (✓)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?	YES ()	NO (✓)
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?	YES ()	NO (✓)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.	YES ()	NO (✓)
E. Name of Regulatory body or any other approvals if required. I. Drugs Controller General of India (DCGI) II. Central Drugs Standard Control Organization (CDSCO) III. Indian Council of Medical Research (ICMR) IV. Ethics Committees (ECs) V. Clinical Trials Registry of India (CTRI)	YES (✓)	NO ()

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us. NA

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity). NA

7. Potential Chances of Commercialization for Smart Helmet with Health Monitoring System

The Smart Helmet with Health Monitoring System possesses strong commercialization potential across multiple sectors due to its advanced safety and healthcare features.

The increasing emphasis on road safety has significantly raised the demand for intelligent helmet solutions in the market. This system provides motorcycle riders, industrial workers, and adventure enthusiasts with enhanced protection through integrated vital sign monitoring, accident detection, emergency alert mechanisms, and collision warning features.

The expanding wearable healthcare market further increases the attractiveness of this product. Motorcyclists, delivery personnel, and factory workers show growing interest in safety devices that support proactive health monitoring and real-time health assessment.

Strategic collaborations with road safety authorities and insurance providers can create valuable business opportunities. Consumer adoption may increase through insurance incentives, subsidies, and safety awareness initiatives.

The emergency connectivity feature adds substantial value by enabling rapid medical intervention. Accurate health data and precise GPS-based location tracking significantly reduce emergency response time, making the helmet highly beneficial for security teams and healthcare professionals.

The system's configurable features allow it to meet industrial safety requirements by detecting toxic gases, temperature extremes, and physical impacts, making it suitable for mining operations, manufacturing facilities, and construction environments.

Additionally, outdoor enthusiasts such as bikers and adventure riders are attracted by the helmet's GPS tracking and alert functionalities. Integration with smart city infrastructure and IoT-based traffic management systems further enhances its commercial appeal.

Large-scale delivery and logistics companies can deploy the system to improve workforce safety while complying with occupational safety regulations.

8. List of Companies for Commercialization of Smart Helmet with Health Monitoring System

1. Steelbird Helmets

Steelbird is a leading helmet manufacturer in India that offers innovative helmet designs. They actively collaborate on smart helmet solutions with integrated safety features.

Website: steelbirdhelmet.com

2. SHOEI Co., Ltd.

SHOEI is a global leader in high-performance motorcycle helmets, specializing in integrating safety and comfort features, making them ideal for smart helmet collaborations.

Website: shoei.com

3. LS2 Helmets

LS2 offers technologically advanced helmets with Bluetooth connectivity, making them a potential partner for smart helmet innovations.

Website: ls2helmets.com

4. AGV Helmets (Dainese Group)

AGV designs advanced motorcycle helmets with enhanced safety and communication systems, making them suitable for smart helmet commercialization.

Website: agv.com

5. Vega Helmets

Vega is an established Indian helmet manufacturer known for developing feature-rich helmets. Their experience can aid in integrating health monitoring features.

Website: vegaauto.com

6. Fastrack Smart Wearables (Titan Company Limited)

Fastrack is a trusted Indian brand in smart wearable solutions, making them ideal for developing health monitoring features in helmets.

Website: fastrack.in

7. Blauer HT Helmets

Blauer HT specializes in developing helmets for safety and security personnel, with potential for integrating health monitoring systems.

Website: blauerht.com

8. Quin Design Helmets

Quin Design Helmets offers smart helmets with crash detection, emergency alerts, and Bluetooth connectivity, making them ideal collaborators for healthcare-focused helmet systems.

Website: quintessential.design

9. Any basic patent which has been used and we need to pay royalty to them. NA

10. FILING OPTIONS: Please indicate the level of your work which can be considered for provisional/ complete/ PCT filings **(Complete)**.

11. KEYWORDS: Please provide right keywords for searching your invention.

1. Smart Helmet Technology
2. Health Monitoring Helmet
3. Intelligent Safety Helmet
4. Accident Detection Helmet
5. Helmet with Vital Sign Monitoring
6. Emergency Alert Helmet System
7. IoT-Based Smart Helmet
8. Wearable Health Monitoring Device
9. AI-Powered Safety Helmet
10. Smart Helmet for Motorcyclists
11. Real-Time Health Monitoring Helmet
12. GPS-Enabled Helmet System
13. Helmet with Heart Rate Monitoring
14. Smart Safety Gear for Riders
15. Industrial Worker Safety Helmet
16. Wireless Health Tracking Helmet

17. Smart Helmet with SOS Feature
18. Bluetooth-Enabled Safety Helmet
19. Sensor-Integrated Helmet System
20. Smart Helmet for Construction Sites